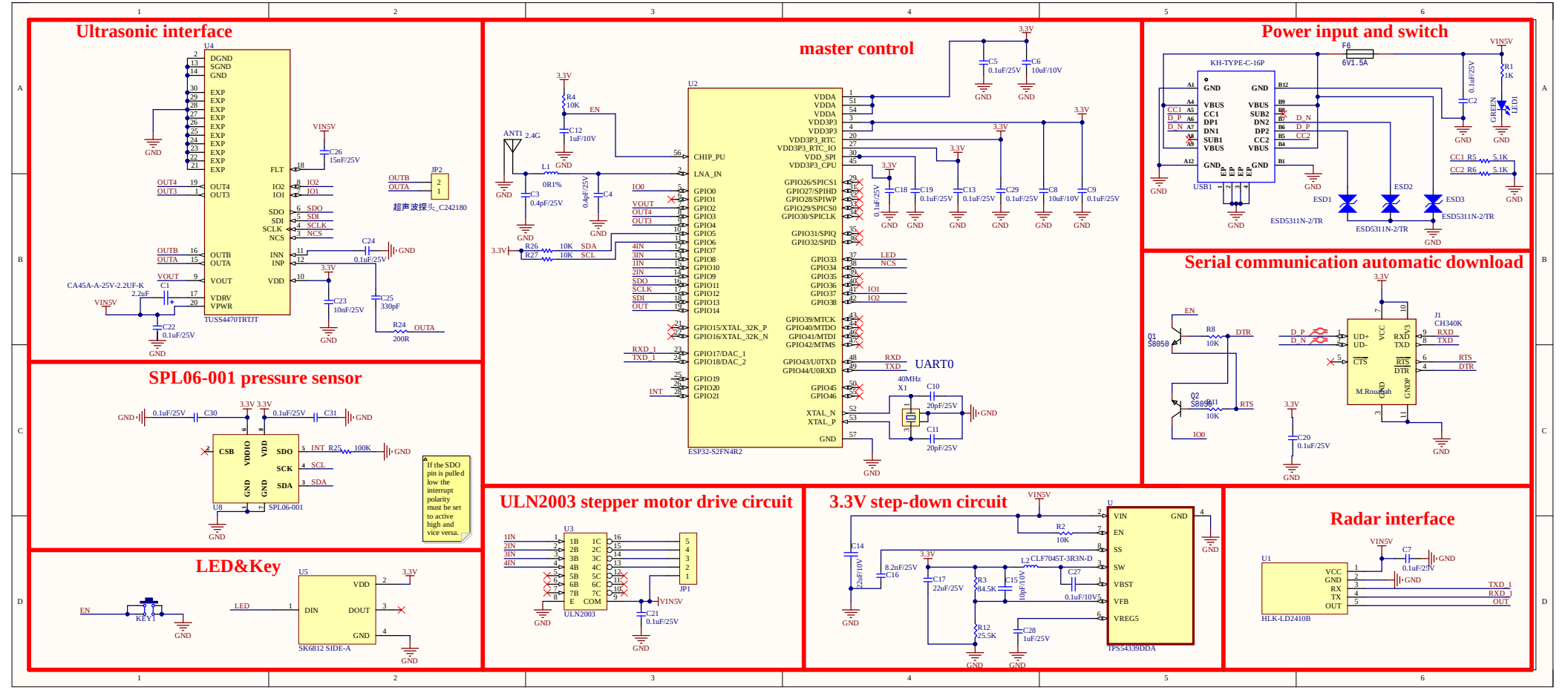
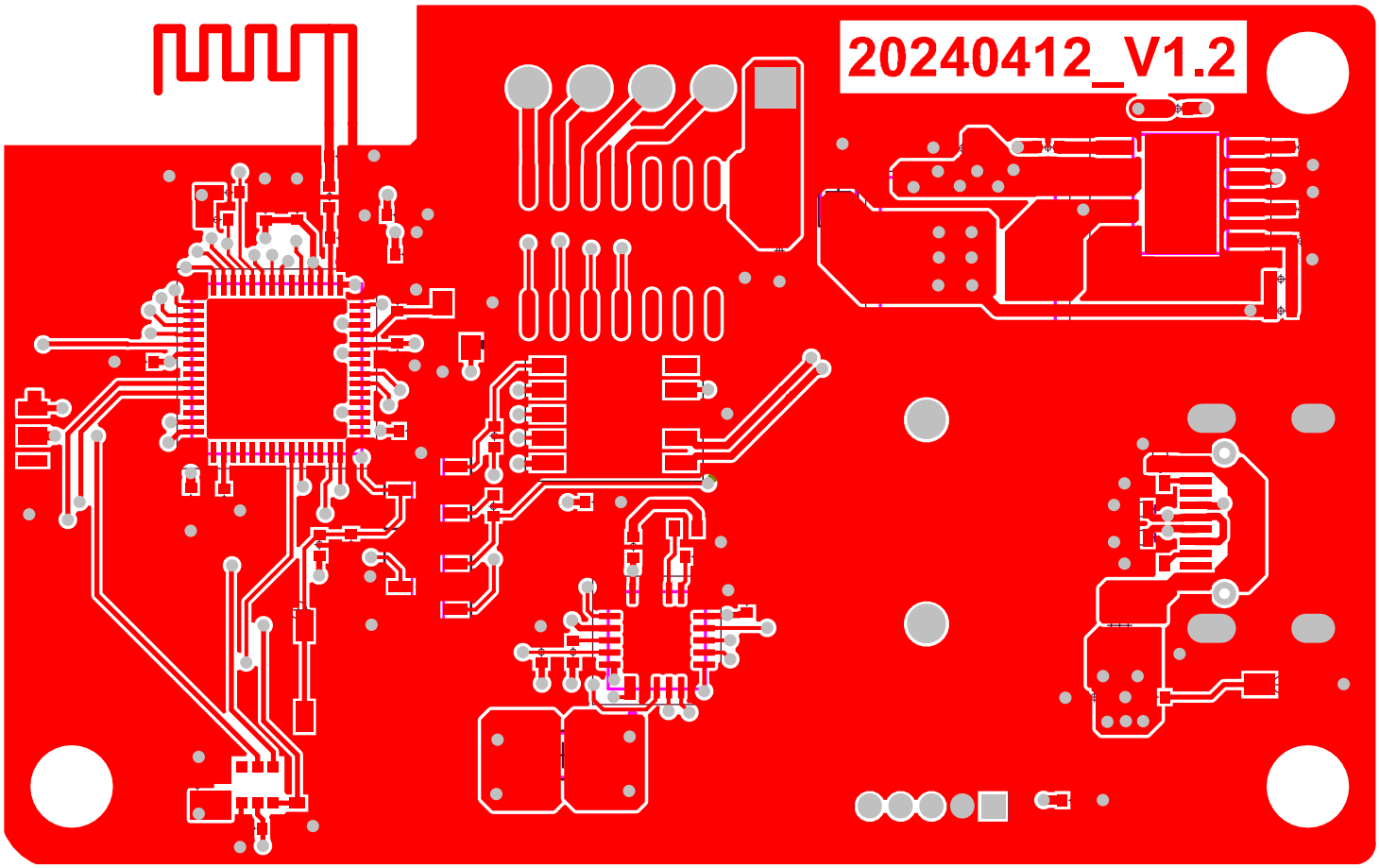
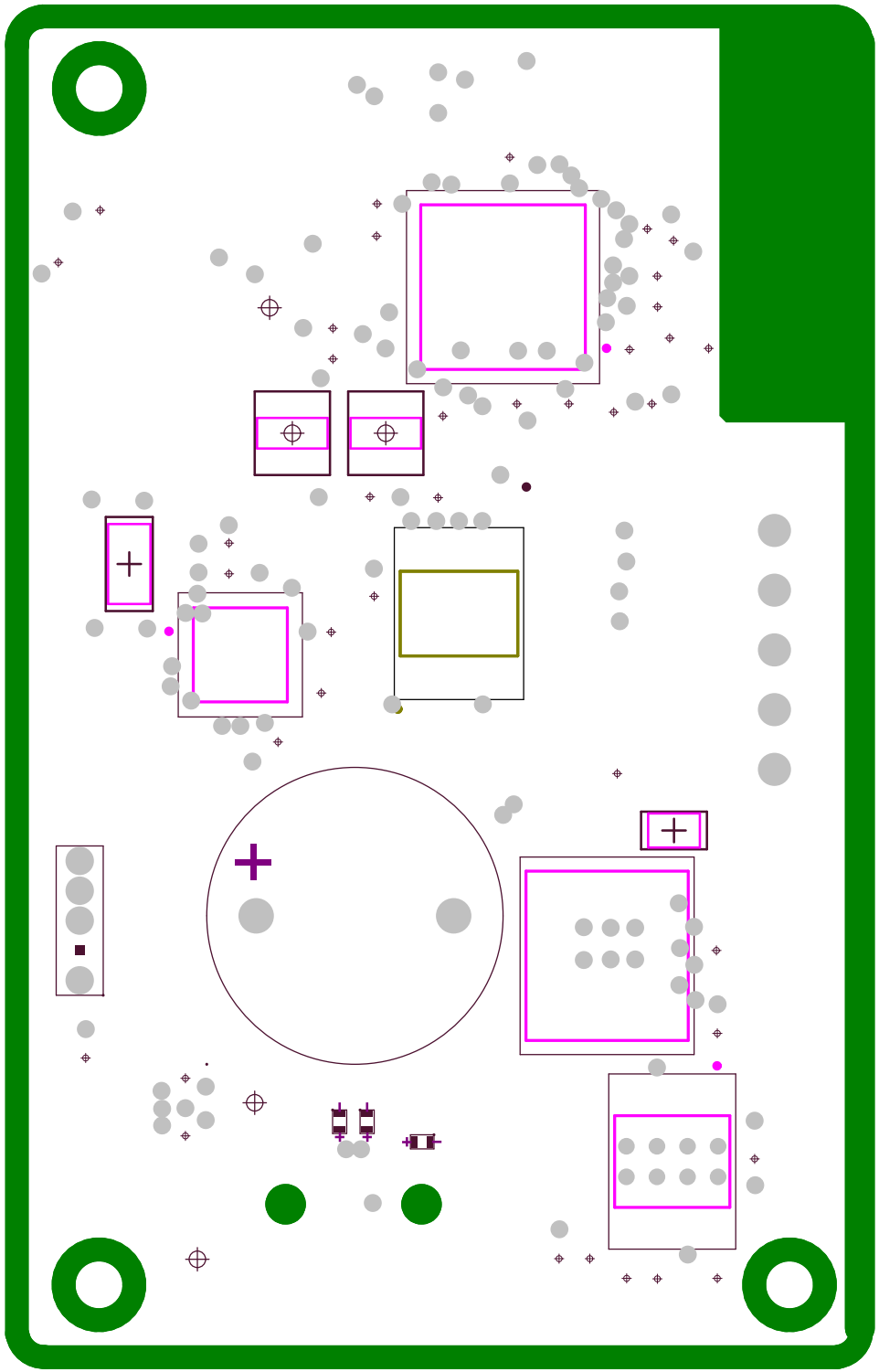
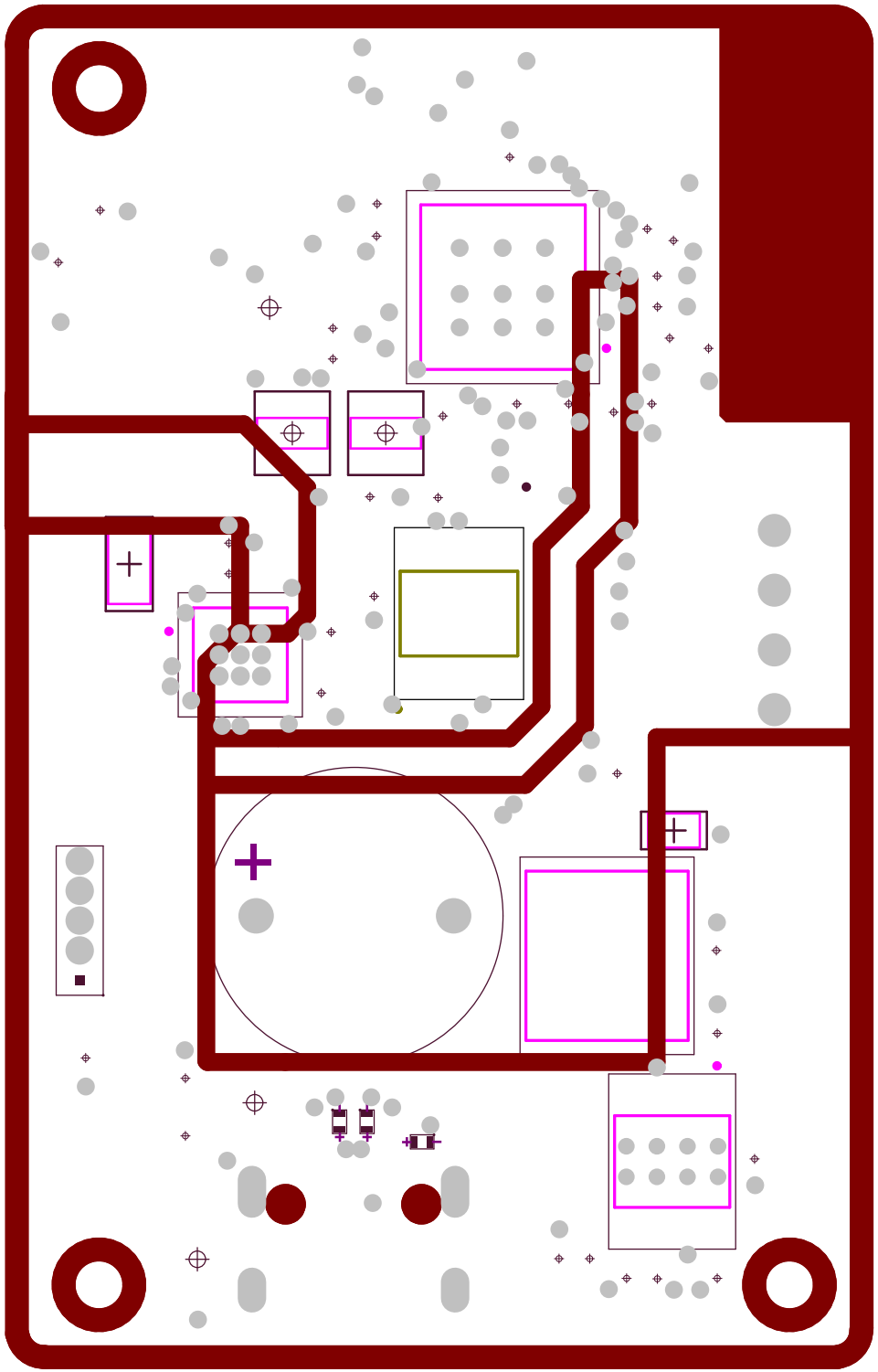
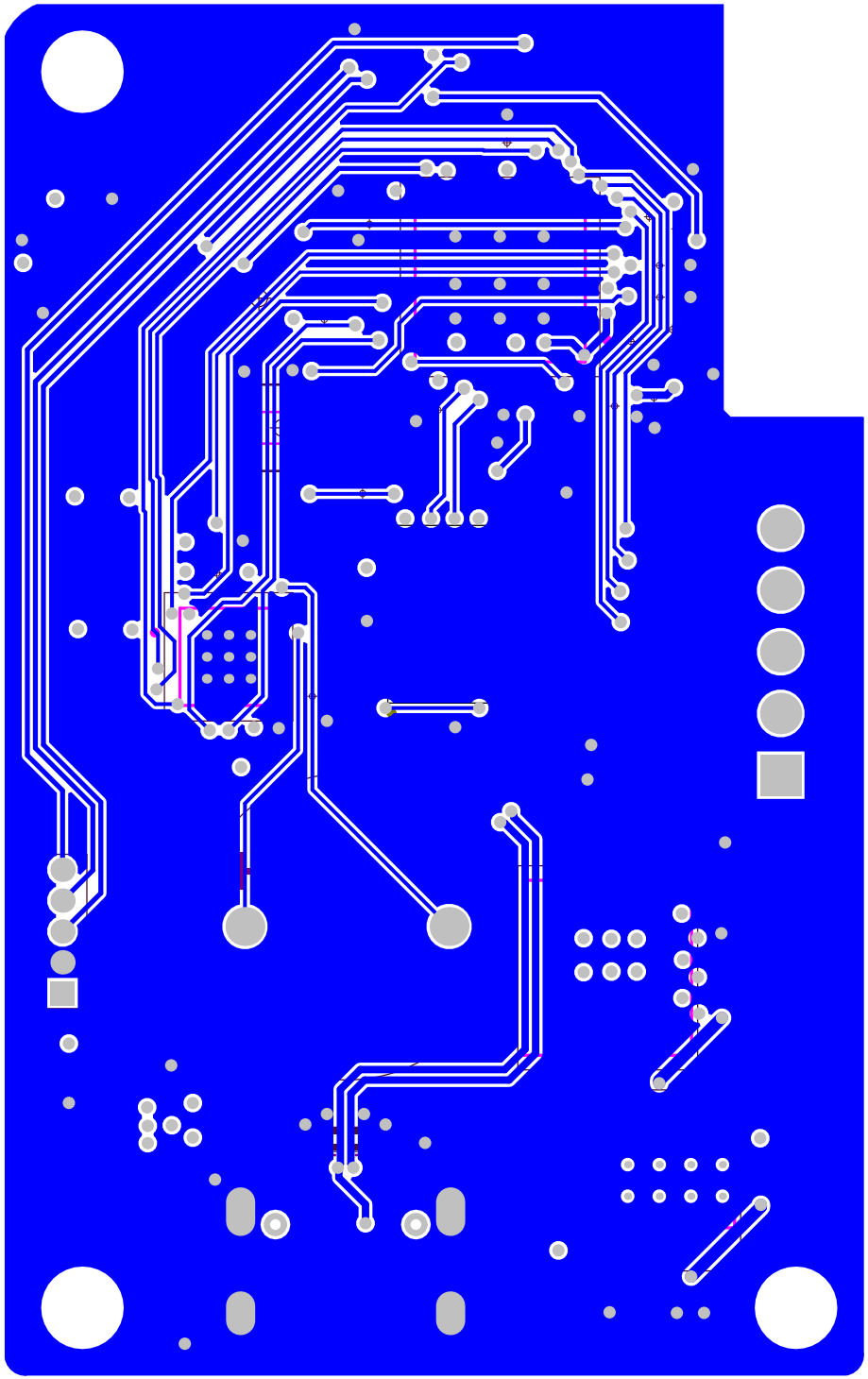
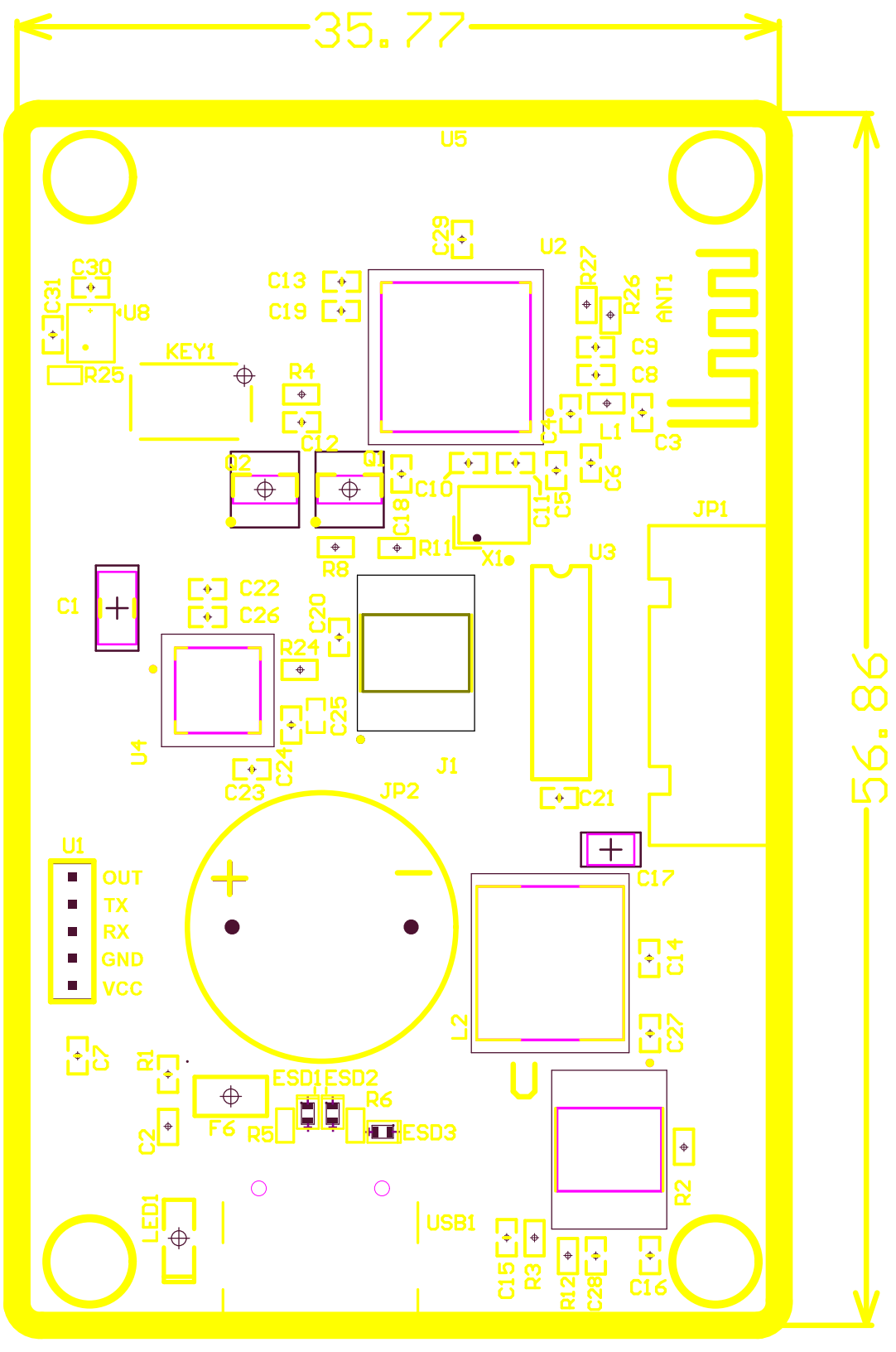

Ultrasonic interface
U4
DGND
SGND
GND
EXP
FLT
IO2
IO1
SDO
SDI
SCLK
NCS
OUT4
OUT3
OUTB
OUTA
INN
INP
VOUT
VDD
VDRV
VPWR
TUSS4470TRTJT
VIN5V
C26
15nF/25V
C24
0.1uF/25V
C23
10nF/25V
C25
330pF
C22
0.1uF/25V
C1
2.2uF
CA45A-A-25V-2.2UF-K
3.3V
GND
JP2
超声波探头_C242180
R24
200R
SPL06-001 pressure sensor
U8
SPL06-001
CSB
VDDIO
VDD
SDO
SCK
SDA
INT
C30
C31
0.1uF/25V
100K
R25
If the SDO pin is pulled low the interrupt mast be set to active high and vice versa.
LED&Key
KEY1
EN
LED
U5
SK6812 SIDE-A
DIN
DOUT
master control
U2
ESP32-S2FN4R2
CHIP_PU
LNA_IN
VDDA
VDD3P3
VDD3P3_RTC
VDD3P3_RTC_IO
VDD_SPI
VDD3P3_CPU
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15/XTAL_32K_P
GPIO16/XTAL_32K_N
GPIO17/DAC_1
GPIO18/DAC_2
GPIO19
GPIO20
GPIO21
GPIO26/SPICS1
GPIO27/SPIHD
GPIO28/SPIWP
GPIO29/SPICS0
GPIO30/SPICLK
GPIO31/SPIQ
GPIO32/SPID
GPIO33
GPIO34
GPIO35
GPIO36
GPIO37
GPIO38
GPIO39/MTCK
GPIO40/MTDO
GPIO41/MTDI
GPIO42/MTMS
GPIO43/U0TXD
GPIO44/U0RXD
GPIO45
GPIO46
XTAL_N
XTAL_P
ANT1 2.4G
R4
10K
EN
C12
1uF/10V
L1
0R1%
C3
0.4pF/25V
C4
0.4pF/25V
R26
R27
10K
SDA
SCL
LED
NCS
IO1
IO2
RXD
TXD
UART0
40MHz
X1
C10
20pF/25V
C11
20pF/25V
C5
0.1uF/25V
C6
10uF/10V
C18
C19
C13
C29
C8
C9
0.1uF/25V
10uF/10V
ULN2003 stepper motor drive circuit
U3
ULN2003
1IN
2IN
3IN
4IN
1B
2B
3B
4B
5B
6B
7B
E
1C
2C
3C
4C
5C
6C
7C
COM
JP1
C21
0.1uF/25V
3.3V step-down circuit
U
TPS54339DDA
VIN
EN
SS
SW
VBST
VFB
VREG5
GND
R2
10K
C14
22uF/10V
C16
8.2nF/25V
C17
22uF/25V
L2
CLF7045T-3R3N-D
R3
84.5K
C15
10pF/10V
C27
0.1uF/10V
R12
25.5K
C28
1uF/25V
Power input and switch
KH-TYPE-C-16P
GND
VBUS
CC1
DP1
DN1
SUB1
SUB2
DN2
DP2
CC2
F6
6V1.5A
VIN5V
R1
1K
C2
0.1uF/25V
GREEN
LED1
CC1_R5
5.1K
CC2_R6
5.1K
USB1
ESD1
ESD2
ESD3
ESD5311N-2/TR
Serial communication automatic download
Q1
S8050
Q2
S8050
R8
10K
R11
10K
DTR
RTS
IO0
EN
J1
CH340K
UD+
UD-
VCC
V3
RXD
TXD
CTS
RTS
DTR
GND
M.Rouqash
C20
0.1uF/25V
Radar interface
U1
HLK-LD2410B
VCC
GND
RX
TX
OUT
C7
0.1uF/25V
TXD_1
RXD_1
OUT


20240412_V1.2








35.77
56.86
U5
C29
U2
C30
C31
U8
C13
C19
R27
R26
ANT1
C9
C8
C3
KEY1
R4
C12
Q2
Q1
C10
C4
L1
C6
C11
C5
C18
R11
X1
R8
JP1
U3
C1
C22
C26
C20
R24
C25
U4
C24
C23
J1
C21
JP2
U1
OUT
TX
RX
GND
VCC
C17
C14
L2
C27
C7
R1
ESD1
ESD2
R6
F6
R5
ESD3
C2
R2
LED1
USB1
C15
R3
R12
C28
C16

1C
2C
3C
4C
+5V

